Bohan Zhang

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Academic Profile

Computer systems and software engineering for trustworthy autonomous systems, with emphasis on computer architecture and systems preparation, high-fidelity simulation, automated testing, requirements engineering, artificial intelligence, LLM-agent testing, and human-reviewed assurance. Eight peer-reviewed publications, including four first-author papers.

Education

Degree: Ph.D. in Computer Science, expected May 2027
Institution: Saint Louis University
Location: St. Louis, Missouri
Date: Fall 2022 to Present

- Ph.D. candidacy examination passed in 2025. Graduate GPA: 3.95/4.00.
- Dissertation draft: *Advancing High-Fidelity sUAS Simulation for Autonomous System Validation*.
- Advisor: Dr. Ankit Agrawal. Ph.D. student researcher supported through the NASA University Leadership Initiative.

Degree: M.S. in Computer Science, non-thesis, GPA: 3.95/4.00
Institution: Saint Louis University
Location: St. Louis, Missouri
Date: May 18, 2024

Degree: B.S. in Computer Science; Minor in Mathematics; official GPA: 3.43/4.00
Institution: Saint Louis University
Location: St. Louis, Missouri
Date: June 23, 2021

Research and Professional Experience

Job Title: Teaching Assistant
Employer: Saint Louis University, Department of Computer Science
Location: St. Louis, Missouri
Dates: September 2026 to December 2026

Job Description

- Support instructional delivery for CSCI 4750/5750, Deep Learning.
- Prepare homework assignments and grade submitted homework.

Job Title: Doctoral Research Assistant
Employer: UAVLab, Saint Louis University
Location: St. Louis, Missouri
Dates: Fall 2022 to August 31, 2026
Supervisor: Dr. Ankit Agrawal

Job Description

- Initiated and developed DroneReqValidator/DroneWorld, a requirements-driven sUAS simulation-testing ecosystem that configures realistic environments, monitors safety properties, and produces acceptance reports. Continued open-source mentoring through OSS-SLU.
- Designed DroneWiS, integrating OpenFOAM and Unreal Engine 5 for physics-based wind simulation around terrain and structures.
- Developed React-G, a GNSS multipath simulation framework using Unreal Engine ray casting, Google Maps 3D Tiles, and Cesium for Unreal. Validated against NASA NavQ measurements.
- Co-developed AutoSimTest, a multi-agent LLM framework using retrieval-augmented generation and structured agents for scenario generation, environment configuration, mission planning, and flight-log analysis.
- Led an NSF-sponsored I-Corps customer-discovery initiative with 22 interviews of robotics and aerospace professionals. No personal award payment was received.
- Contributed to SADE-SIM and AutonomyLens, including scalable city-scale multi-sUAS validation and a self-evolving LLM-agent simulation-testing loop.

Job Title: Assistant Engineer Intern
Employer: Beijing VRV Software Corporation Limited, Xi'an Branch
Location: Xi'an, Shaanxi, China
Dates: March 10 to June 30, 2022
Supervisor: Not applicable

Job Description

- Contributed to front-end and back-end development for enterprise software and implemented unit and integration tests for a

web-server backend.
- Developed a secure Java graphical application for file encryption and decryption.

Publications

1. **B. Zhang**, A. Putta, and A. Agrawal. Advancing Small Unmanned Aerial Systems Simulation Testing in Realistic Windy Conditions. *IEEE Access*, 2026. DOI: 10.1109/ACCESS.2026.11482137.
2. A. Agrawal, J. Garapati, and **B. Zhang**. AutonomyLens: A Self-Evolving Simulation-Based Testing Loop for Autonomous Systems. *2nd International Workshop on Software Engineering for Engineering Simulations, co-located with FSE*, 2026. arXiv:2604.11672.
3. L. Vadnala, L. Parzianello, **B. Zhang**, M. Murphy, A. Agrawal, and D. Thain. SADE-SIM: A Scalable Simulation Platform for Validating City-Scale Multi-sUAS Missions. *SE4ES, co-located with FSE*, 2026. DOI: 10.1145/3803437.3805540.
4. V. S. A. Duvvuru, **B. Zhang**, M. Vierhauser, and A. Agrawal. LLM-Agents Driven Automated Simulation Testing and Analysis of Small Uncrewed Aerial Systems. *47th IEEE/ACM International Conference on Software Engineering*, pp. 385-397, 2025. DOI: 10.1109/ICSE55347.2025.00223.
5. **B. Zhang**, J. Gutierrez, and A. Agrawal. React-G: GNSS Signal Multipath Simulation Framework for Small Uncrewed Aerial Systems. *IEEE/ION Position, Location and Navigation Symposium*, pp. 1170-1181, 2025. DOI: 10.1109/PLANS60068.2025.11028333.
6. **B. Zhang** and A. Agrawal. DroneWiS: Automated Simulation Testing of Small Unmanned Aerial System in Realistic Windy Conditions. *39th IEEE/ACM International Conference on Automated Software Engineering*, pp. 2358-2361, 2024. DOI: 10.1145/3691620.3695351.
7. **B. Zhang**, Y. Shivalingaiah, and A. Agrawal. DroneReqValidator: Facilitating High Fidelity Simulation Testing for Uncrewed Aerial Systems Developers. *38th IEEE/ACM International Conference on Automated Software Engineering*, pp. 2082-2085, 2023. DOI: 10.1109/ASE56229.2023.00011.
8. A. Agrawal, **B. Zhang**, Y. Shivalingaiah, M. Vierhauser, and J. Cleland-Huang. A Requirements-Driven Platform for Validating Field Operations of Small Uncrewed Aerial Vehicles. *31st IEEE International Requirements Engineering Conference*, pp. 29-40, 2023. DOI: 10.1109/RE57278.2023.00013.

Invited Teaching

Role: Invited Guest Lecturer and Workshop Leader
Institution: Saint Louis University, Department of Computer Science
Location: St. Louis, Missouri
Dates: Spring 2025 to Spring 2026

- CSCI 5030, Principles of Software Development, Spring 2025: Git and GitHub, branches, pull requests, code review, merge workflows, and conflict resolution.
- CSCI 4930/5930, Human-Computer Interaction, Spring 2025: two lectures on sUAS simulation fidelity, the sim-to-reality gap, and human-in-the-loop testing.
- CSCI 3300/5300, Software Engineering, January to May 2026: workshops on test-driven development, unit testing, merge workflows, Docker, containerization, and continuous integration with GitHub Actions.
- Spring 2026 evaluation, 10 respondents: hands-on impact 4.80/5; instructor effectiveness 4.70/5; recommendation to invite again 4.90/5.

Teaching interests: Computer Organization and Design; Computer Architecture; Digital Logic; Operating Systems; Software Engineering; Software Testing; Artificial Intelligence; Machine Learning; Computer Security; Introductory Programming; Data Structures and Algorithms; Computer Networks; Human-Computer Interaction; Capstone and research-project supervision.

Selected Presentations

- React-G, IEEE/ION PLANS, Salt Lake City, Utah, April 28, 2025.
- DroneWiS, ASE, Sacramento, California, October 27, 2024.
- RealWindDroneSim poster, NASA imaginAviation University Poster Session, February 28, 2024.
- DroneReqValidator, ASE, Echternach, Luxembourg, November 11, 2023.

Professional Service

- Peer Reviewer, IEEE International Conference on Robotics and Automation, 2026.
- Reviewer, *IEEE Software* special issue on software for small uncrewed aerial systems, 2026.

Honors and Awards

- Alpha Sigma Nu Honor Society, Fall 2024.
- NASA imaginAviation University Poster Individual Award, 5th Place, Spring 2024.
- Global Graduate Scholar Award, Saint Louis University, Fall 2022 to Spring 2024, $10,000.
- James G. Costigan Endowed Scholarship, Saint Louis University, Spring 2021, $5,000.

Technical Skills and Languages

Simulation and engines: Unreal Engine 5, AirSim, Gazebo, OpenFOAM.
Programming: Python, JavaScript, Java, C++.
Development: Docker, GitHub Actions, React, Flask, REST APIs, unit and integration testing.

Methods: Computational fluid dynamics, ray casting, voxelization, retrieval-augmented generation, LLM-agent orchestration, runtime monitoring.
Languages: English and Chinese, native/bilingual; Japanese, elementary.